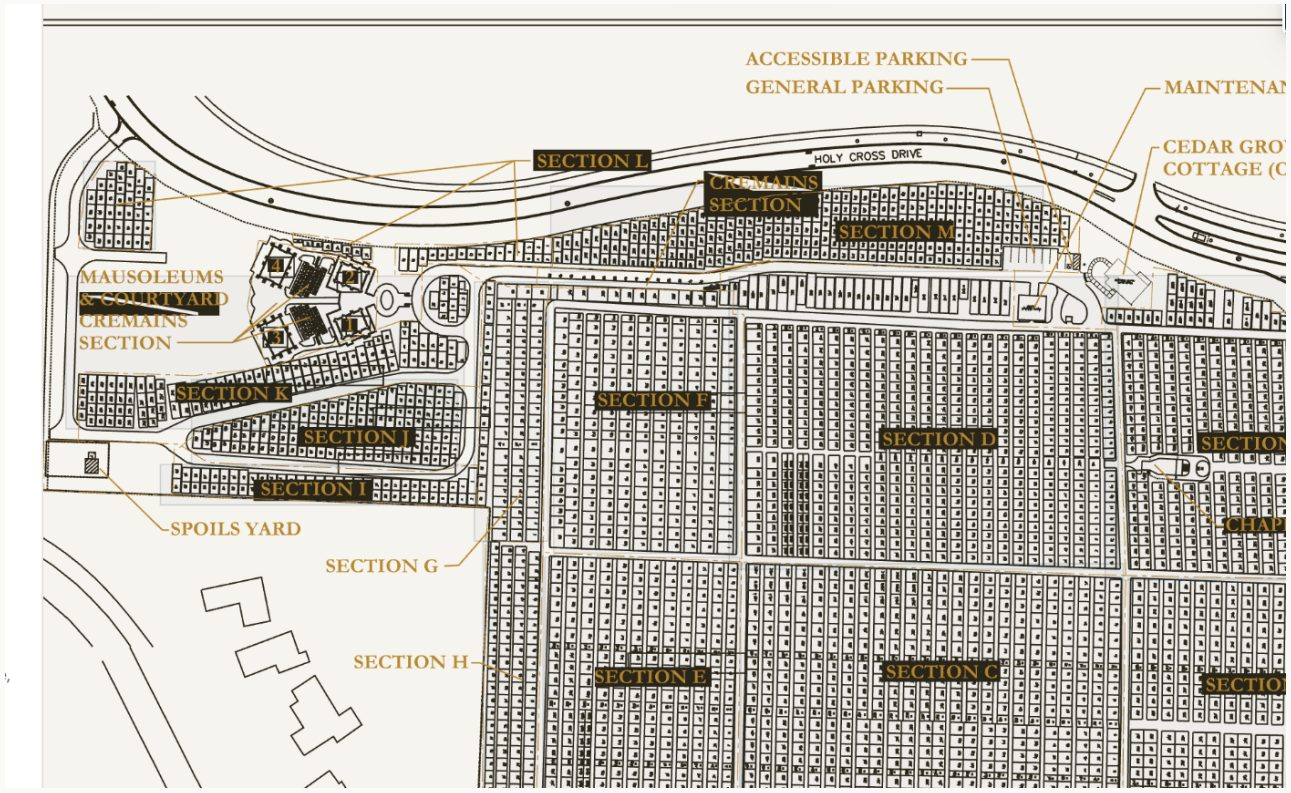
From paper plans to a working atlas

A visual explanation of the mapping system, its current limits and the information needed to complete the grave records.

THE PURPOSE

One place to find, understand and print cemetery locations.

Translate legacy labels into H-numbers, locate lots on the drawings and connect verified grave information.



3,063

placed rows reported

136

unplaced entries reported

49

unmatched drawing labels

Decision focus: reconcile numbering conflicts, verify lot capacity and establish an authoritative master dataset. Counts are supplied project figures, not an independent audit.

How the map is produced

The system reads the original vector drawing, connects each label to the decoder and builds reusable map data.

01

READ SHAPES

Group vector strokes into characters; compare rotated shapes with templates.

glyphs.py / recog.py

→

02

ALIGN PAGES

Place the 9 section pages into the overall map coordinate system.

register.py / reg2.py

→

03

ASSIGN SECTION

Use section boundaries to determine where each recognized label belongs.

region.py / assign.py

→

04

DECODE + REVIEW

Translate legacy labels; reconcile candidates and flag unresolved cases.

evalsec.py / finalize.py

→

05

RENDER

Replace labels while preserving their position and rotation.

render.py / build_final.py

Original sources

Overall Map PDF + 9-page section book.

Vector strokes require shape recognition: there is no searchable label text.

→

Shared map dataset

Identity + section + position + angle + review status.

The decoder supplies numbering; geometry supplies location.

→

Two output branches

Print: vector maps and section PDFs.

Website: precomputed data + SVG artwork + shared records.

Why this structure: recognize once, review once, then reuse the same approved identities in the website and printed maps. The site does not rerun recognition.

The mathematics of the H-number

A decoder combines formula-based sweeps with ordered tables. The exact historical tables remain the numbering authority.

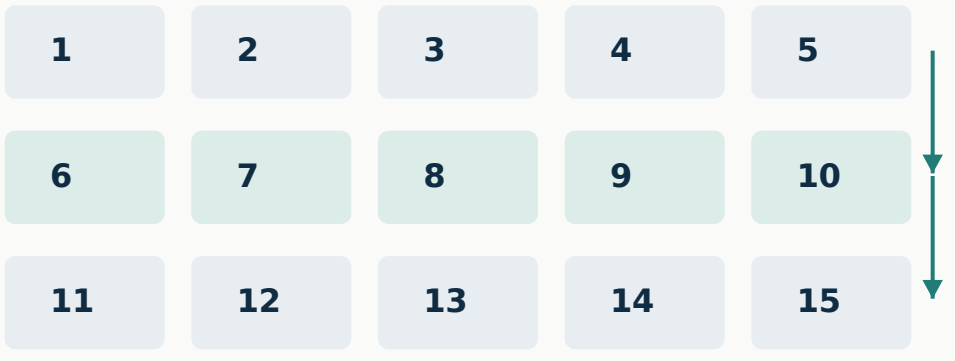
Formula-based sweep

$$H_{\text{lot}} = O + S(r - 1) + c$$

O = offset before this sweep; S = sweep width.
r = row index; c = position within the sweep.

Reported main-grid rules: B uses O = 0; A uses O = N_B. Margin lookups can override these rules.

Reading order → numbering order



Illustrative sweep only: five positions per row. Actual section paths and widths come from the decoder.

Table-based continuation

$$H_{\text{lot}} = O_s + \text{rank}_{T_s}(\ell)$$

T_s is the approved ordered lot list; rank starts at 1. Each sweep continues the running H-number counter.

Reported strategy families
A/B: generated label-to-row/column maps.
C/D and E/F: ordered tables and row suffixes.
Margins, G/H, M and K/J/I/L: continuation tables.

Source limitation: these equations summarize the supplied notes. Full constants, branches and row tables require the actual decoder file before certification.

A grave number is not a burial record

The code reserves positions; it does not tell us how many graves a lot actually contains or whether someone is buried there.

$$H_{\text{grave}} = 10H_{\text{lot}} + p$$

Existing formula: p ∈ {1, …, 9}. This is a numbering envelope, not a verified capacity of nine graves per lot.

LOT

H001871

Illustrative lot identifier

→

POSITION

3

A selected position

→

GRAVE ID

H018713

10 × 1871 + 3

Capacity must be verified

Lots may contain different numbers of graves, including 1-5, 1-6 or 1-7 as described. Record unknown capacity as unknown; do not assume a uniform count.

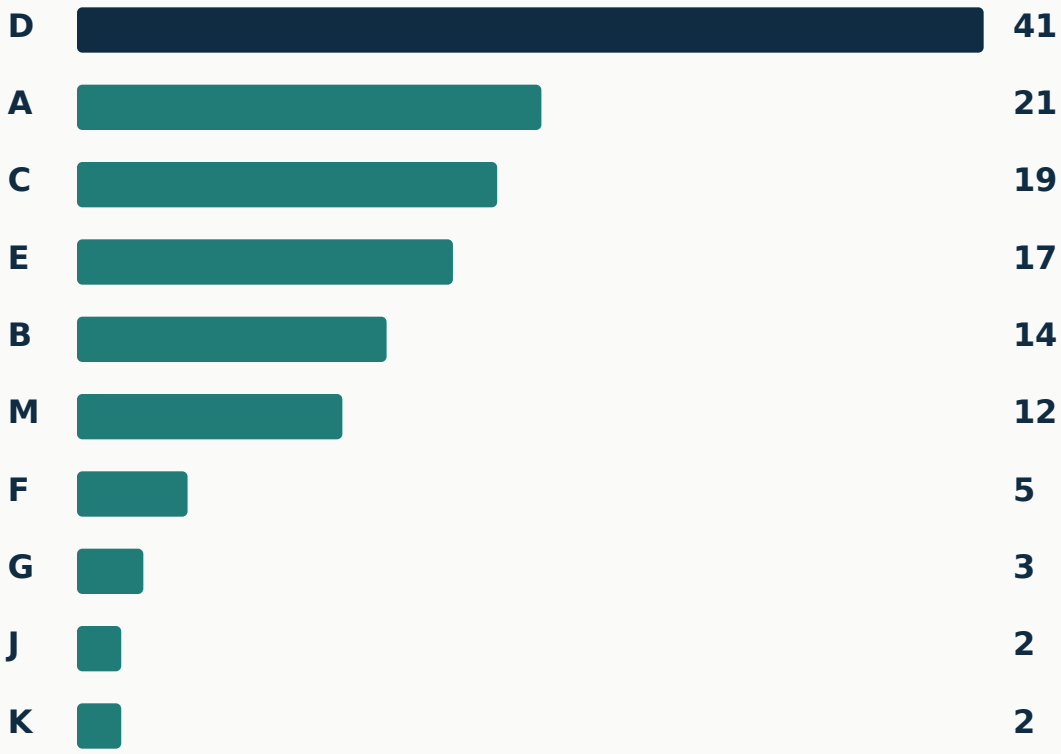
Store valid positions explicitly

Use an approved position set P for each lot. A verified grave position requires p ∈ P. Count = |P|. Use 1…n only if consecutive numbering is confirmed.

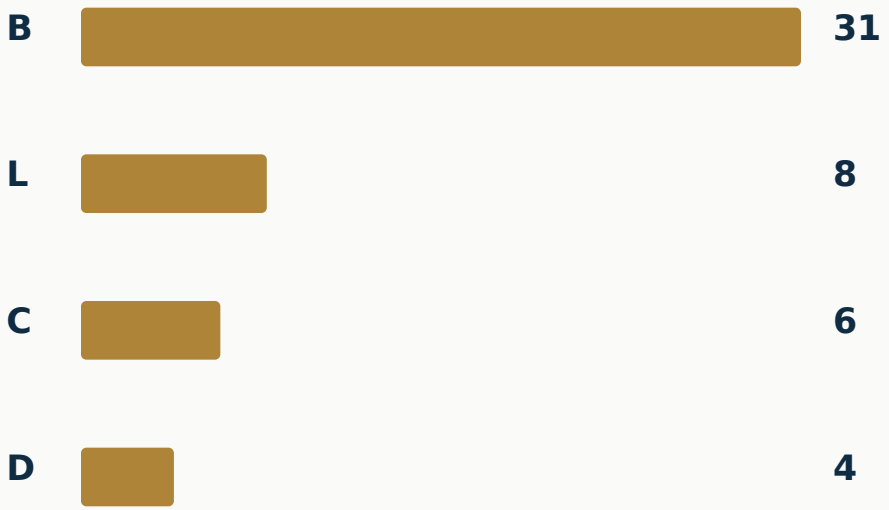
Where the review work is concentrated

Reported unresolved entries by section. These charts show workload, not burial counts or positional accuracy.

136 unplaced catalog entries



49 unmatched drawing labels



H, I and L have zero unplaced entries in the supplied table. L still has 8 unmatched drawing labels: the categories measure different directions.

Reconcile the denominator before publishing a percentage.
3,063 + 136 = 3,199 entries; the stated range H000001-H003185 contains 3,185 possible lot IDs. Notes also describe ~30 bookkeeping/shared-base entries. Unique physical lots must be audited first.

Four issues need targeted decisions

The notes describe plausible causes. Resolve each against source drawings and the numbering tables before changing permanent identities.

A

Suffix disagreement

18 of 21 missing entries follow 20H, 39H, 58H … 343H; the notes report plain numbers on the drawing.

Decision / next action
Review the series; approve aliases only where the physical lot is unambiguous.

B

Two areas share a label

Reported example: B 380H resolves forward to H002341, while reverse lookup of H000793 also returns B 380H.

Decision / next action
Add area-level identity or an approved disambiguator. Do not let lookup order decide location.

G

Duplicate lot 19

The supplied row list repeats 19; the second occurrence consumes an H-number that is not forward-reachable.

Decision / next action
Check the physical drawing. Preserve issued IDs; assess downstream renumbering before removing entries.

ALL

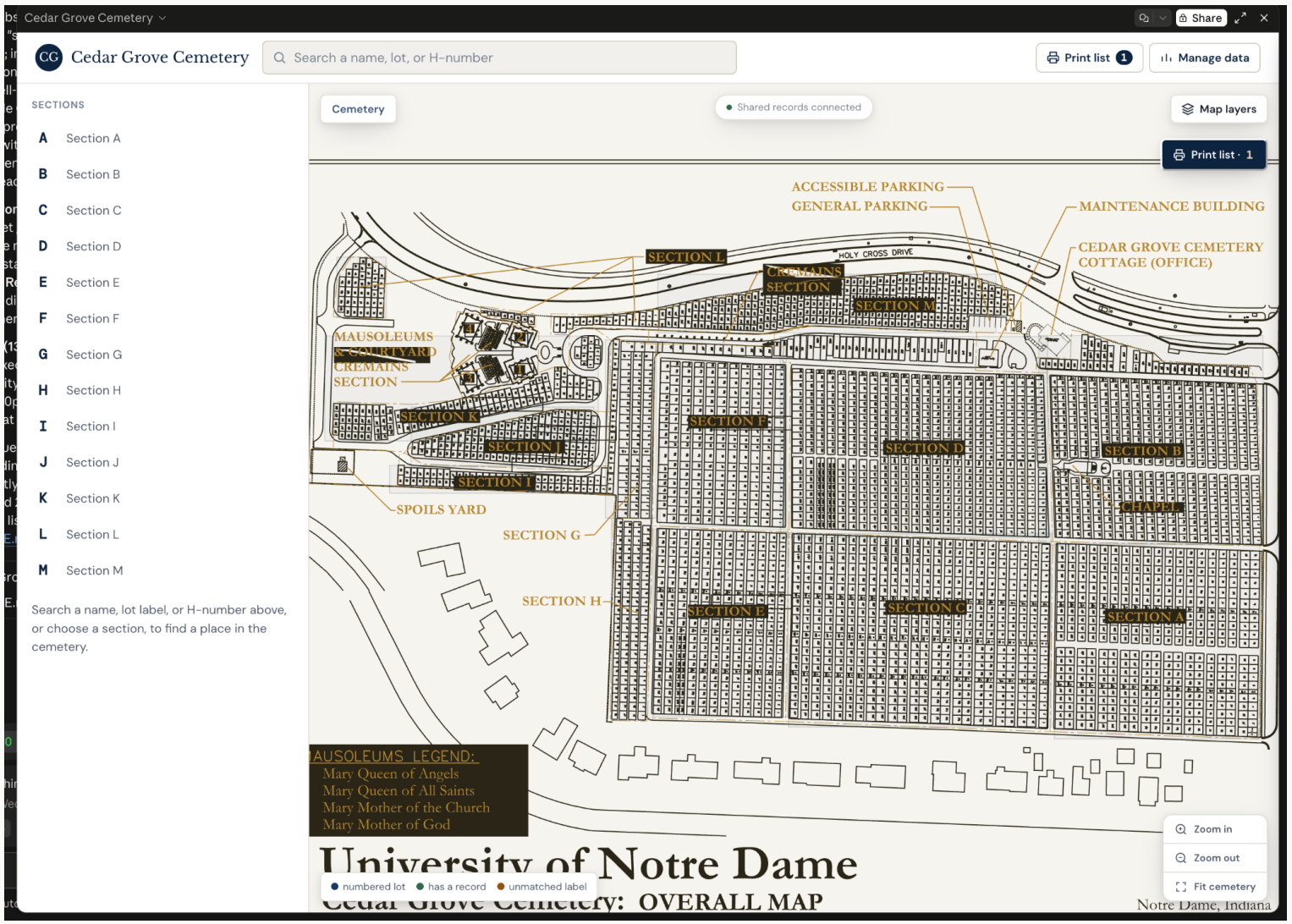
Unresolved map evidence

49 drawing labels lack decoder matches. ~30 of the 136 unplaced entries are described as bookkeeping cases.

Decision / next action
Separate aliases, true missing lots and uncertain recognition; retain a review history.

The website makes the numbering usable

The supplied screenshot illustrates the interface. Current v5 functionality is described in the notes; it was not independently exercised.



01

Search → locate

Find by legacy label, H-number or recorded name. Selection should center and highlight the matching location.

02

Section → lot → grave

Click a section to fit its bounds. Click a lot to zoom again. Show grave details; use lot location when grave coordinates are unknown.

03

Read → edit → save

Keep viewing simple. Save verified record changes to the shared dataset, with clear errors and unsaved-change protection.

Next usability checks: fit the initial map • keep one Print list button • readable labels by zoom level • obvious Back to section • visible section exports.

One master dataset, connected records

A curated dataset connects the numbering system to real grave information and keeps the website and exports consistent.

LOTS

Key: stable lot ID + lot H-number
Section; legacy label; area / alias
Verified capacity or unknown
Approved grave-position set
Map coordinates + rotation
Source document / page
Verification status + reviewer

GRAVE POSITIONS

Key: lot ID + position
Grave H-number
Position verification status
Physical location, if verified
Notes + evidence reference

Keep proposed slots separate from verified positions.

BURIAL RECORDS

Key: stable record ID
Linked grave-position ID
Name; birth; death
Burial date, if available
Notes + register reference
Updated by + timestamp

Allow multiple records if required.

Controlled update loop

IMPORT
CSV / XLSX + column mapping

REVIEW
Exact IDs + conflict preview

COMMIT
Versioned approved records

REFRESH
Site details + new exports

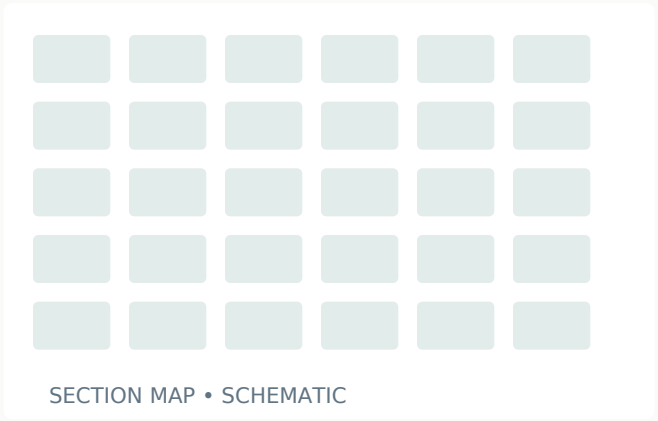
A master workbook can collect the data; the shared database should be the operational source after review. Clicking a grave reads its linked record. Numbering or geometry changes trigger a map rebuild; record edits refresh details and future exports.

Export the section you are working in

Section H and every other section need direct, clearly labeled download actions. Batch export is a next-step requirement.

MAP PDF

A field reference

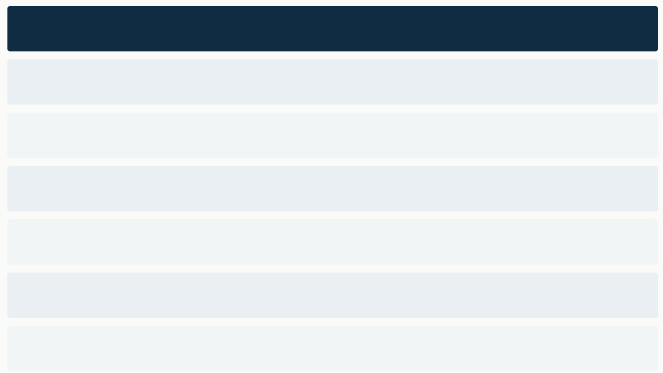


SECTION MAP • SCHEMATIC

Vector section map with H-numbers, orientation and landmarks. Choose sheet size; tile dense sections for readable labels.

DATA PDF

A readable report



Section, legacy lot, lot/grave H-number, recorded names, dates, notes and mapping status. Repeat table headers.

EXCEL .XLSX

A reusable dataset

Sheets: Lots, Grave positions, Burial records, Exceptions and Read me. Preserve H-numbers as text; keep unknown values blank.

All sections: one map PDF per section + a combined indexed atlas; one combined data PDF; one consolidated workbook with a Section column. Include unmapped lots in data exports and disclose them in map exports.

The technical method behind the visuals

These equations explain the algorithms described in the notes. They are mathematical summaries, not a verified transcription of source code.

1 Recognize shape + rotation

$$(\hat{k}, \hat{\theta}) = \arg\max_{k,\theta} \operatorname{corr}(I_\theta, T_k)$$

I = normalized candidate glyph; T = template.
Reported: 36 × 36 blurred grids, 10° coarse sweep, then 2° refinement within ±10°. Alternate readings are retained for review.

2 Register page coordinates

$$\mathbf{q} = \mathbf{A}\mathbf{p} + \mathbf{t}$$

p = section-page point; q = overall-map point.
Initial model: A = sI (scale + translation). The deep-dive describes a later full affine refinement. Preserve label orientation.

3 Improve alignment

$$\min_{\mathbf{A},\mathbf{t}} \sum_i \|\mathbf{q}_i - (\mathbf{A}\mathbf{p}_i + \mathbf{t})\|^2$$

Match reference segments, then minimize alignment error. Reported residual: under 0.05 map units. Its metric and source run need verification; this is not a surveyed ground-distance accuracy.

4 Validate identity + coverage

$$d^{-1}(d(u)) = u$$

$$C = 100\frac{N_{\text{unique mapped lots}}}{N_{\text{unique eligible lots}}}$$

u must be unambiguous. Check forward/reverse identity, duplicate IDs and eligible-lot definitions before reporting C.

What to complete next

Deliver a reproducible atlas first, then expand verified grave records and operational exports.

01 Resolve identity

Review A suffixes, B area collisions and G duplicate. Reconcile aliases and the 3,199 / 3,185 discrepancy.

Cemetery staff + developer

02 Verify grave data

Collect valid positions and actual capacity per lot. Link the burial register with sources and review status.

Cemetery records staff

03 Stabilize the build

Preserve mkdata.py, mksvg.py, site builder and template with the pipeline. Version inputs, decoder and outputs.

Developer

04 Prove the workflow

Test section/lot zoom, record edits and Section H PDF/XLSX exports; then batch all sections and review printed labels.

Developer + staff acceptance

Improve map generation without hiding uncertainty

Build a review queue with cropped source labels, alternate readings and confidence. Expand templates from approved examples; compare against a manually verified sample. Inspect label erasure and line preservation. Do not lower acceptance thresholds merely to inflate coverage.

Evidence & handoff [1] Supplied full project walkthrough, 25 Sep 2026. [2] Attached technical deep-dive, Pasted text(20260925-041912).txt. [3] Supplied interface screenshot, image(6).png, 24 Sep 2026. Exact decoder constants and CSV contents were not provided. Linked code and live artifact could not be retrieved; implementation claims and residuals remain reported, not independently tested.
Required for final audit: decoder + README, both source PDFs, three current CSVs, all build scripts and site source.